Movie Ticket Field Visibility Checker

Practice Problem 1

Concept and approach

Use a modifier/context table for the three contexts. summarizeBatch counts all attempts. MovieTicket's field design: seatNumber private (internal booking consistency), screenId package-private (trusted booking package), ticketPrice protected (subclass loyalty module), and movieTitle public (safe display metadata).

Important implementation notes

For this practice problem, protected outside the package is denied in the generic DIFFERENT_PACKAGE context. The subclass-specific exception is added in Problem 2.

Java solution

```
class AccessChecker {
static String classifyAccess(String modifier, String context) {
switch (modifier) {
case "private": return context.equals("SAME_CLASS") ? "ALLOWED" : "DENIED";
case "default":
case "protected": return context.equals("SAME_CLASS") || context.equals("SAME_PACKAGE") ? "ALLOWED" : "DENIED";
case "public": return "ALLOWED";
default: throw new IllegalArgumentException("Unknown modifier: " + modifier);
}
}

static String summarizeBatch(String[][] attempts) {
int allowed = 0, denied = 0;
for (String[] attempt : attempts) {
if (classifyAccess(attempt[0], attempt[1]).equals("ALLOWED")) allowed++;
else denied++;
}
return "Allowed: " + allowed + " | Denied: " + denied;
}
}

class MovieTicket {
private int seatNumber;
String screenId;
protected double ticketPrice;
public String movieTitle;
}
```

Sample output

```
Allowed: 2 | Denied: 1
```